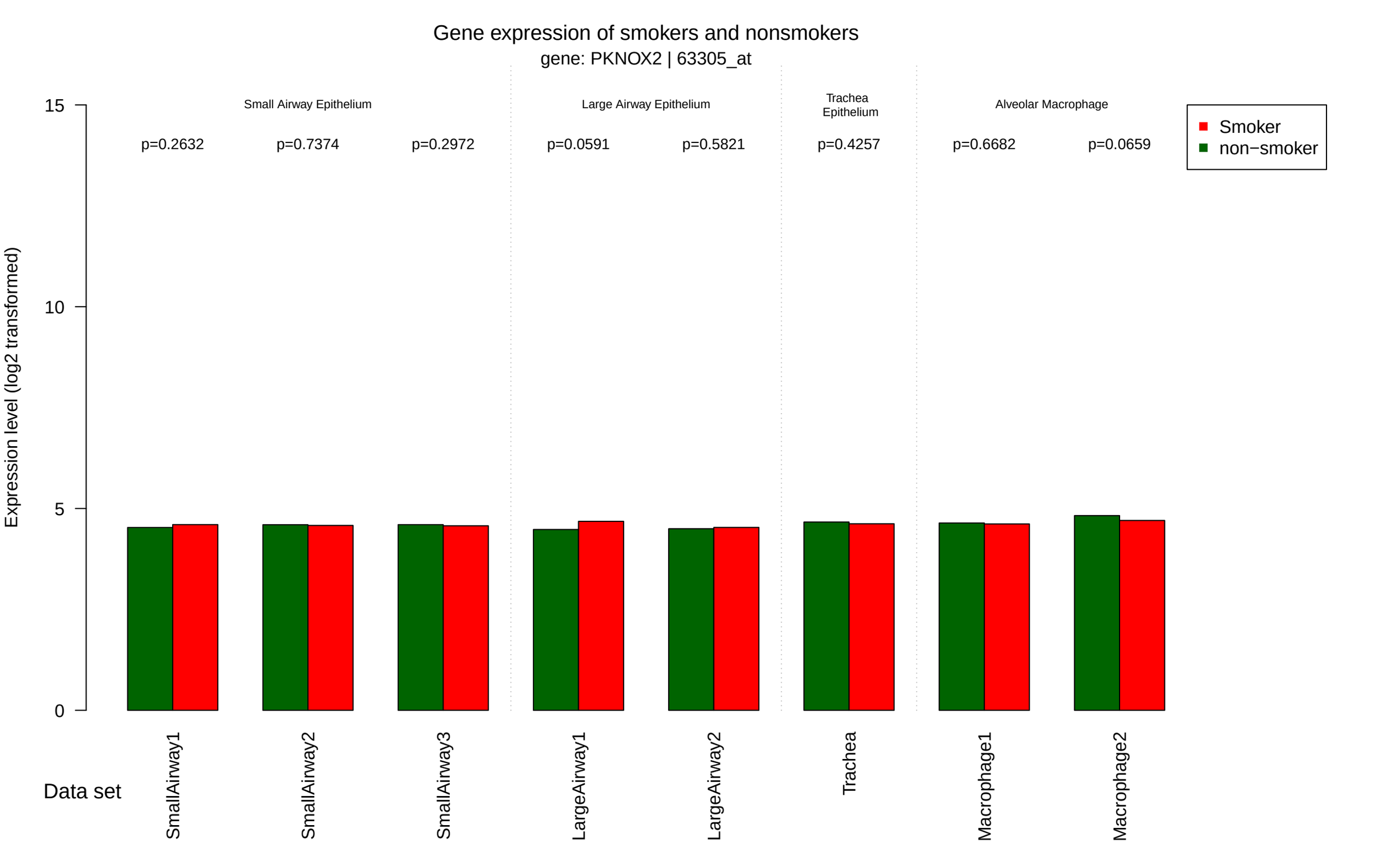
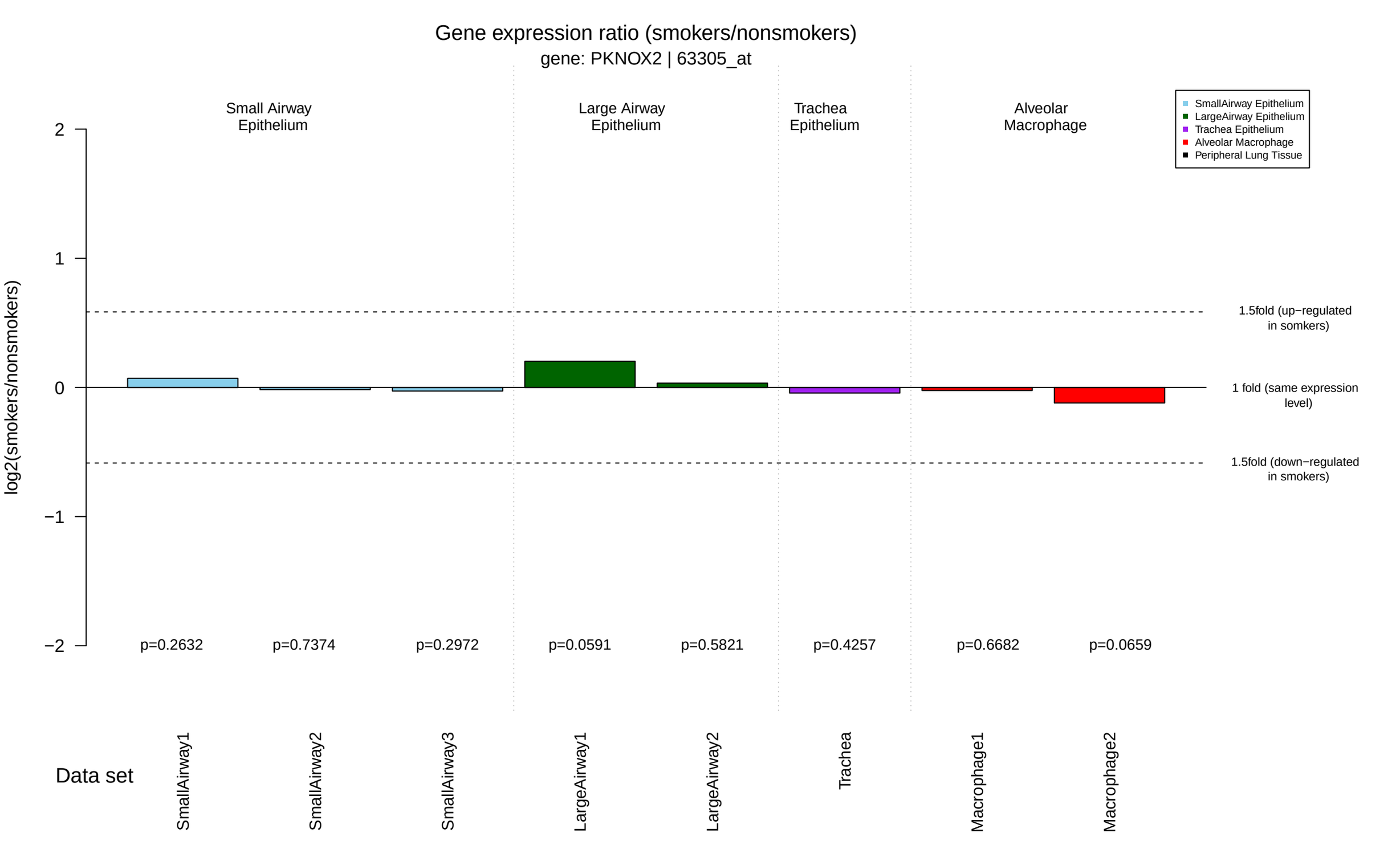
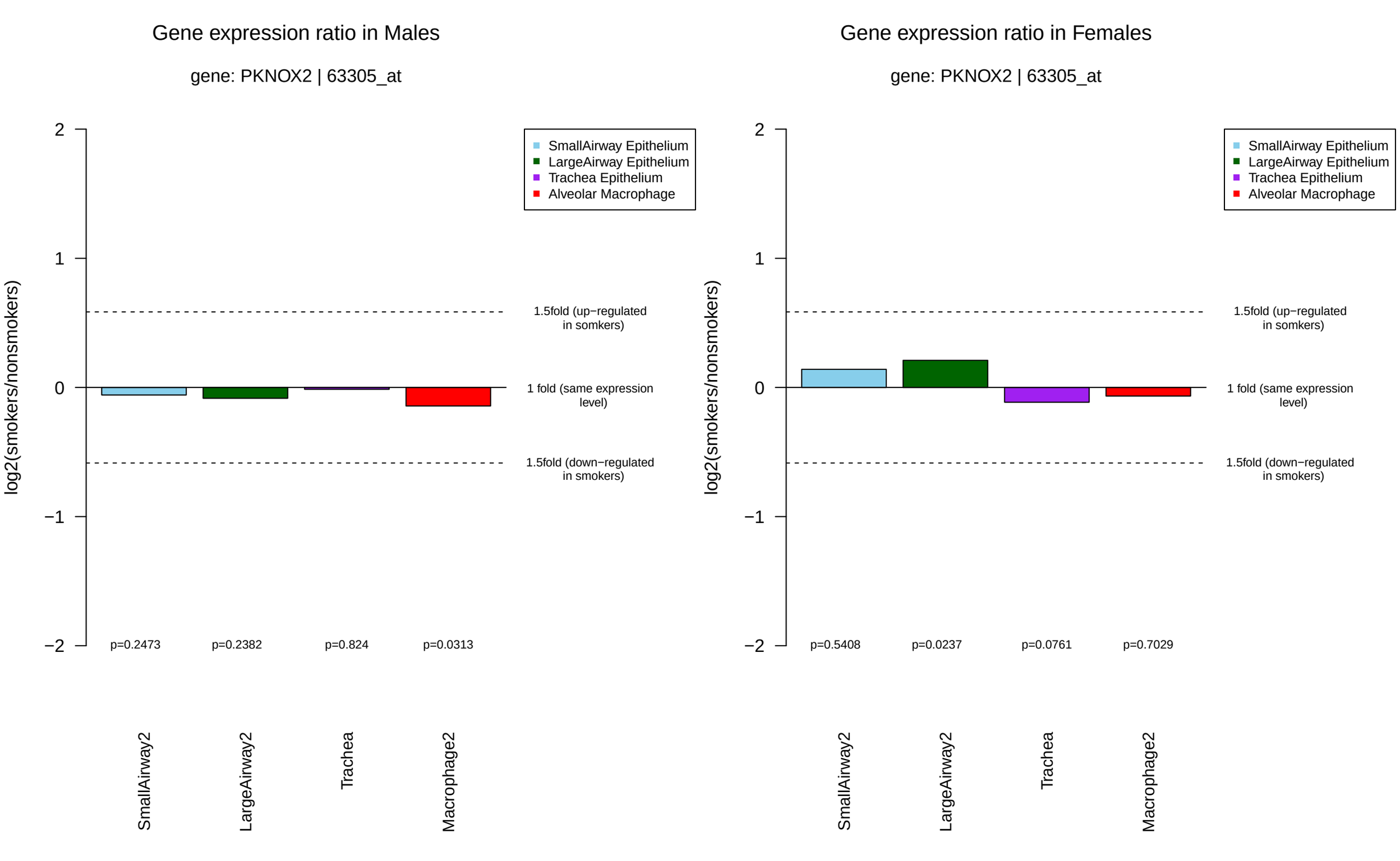
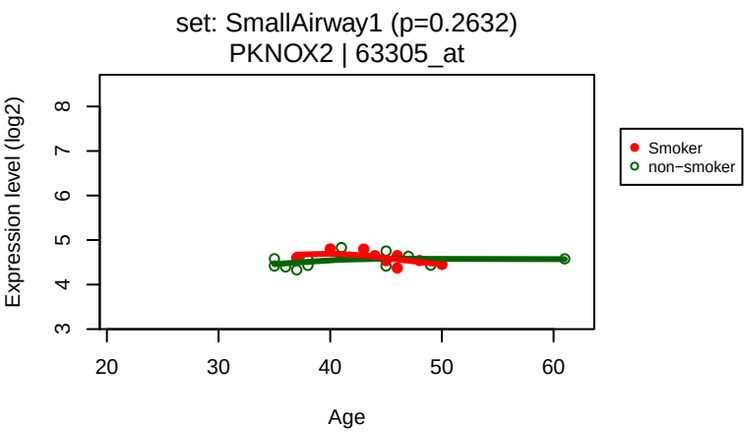
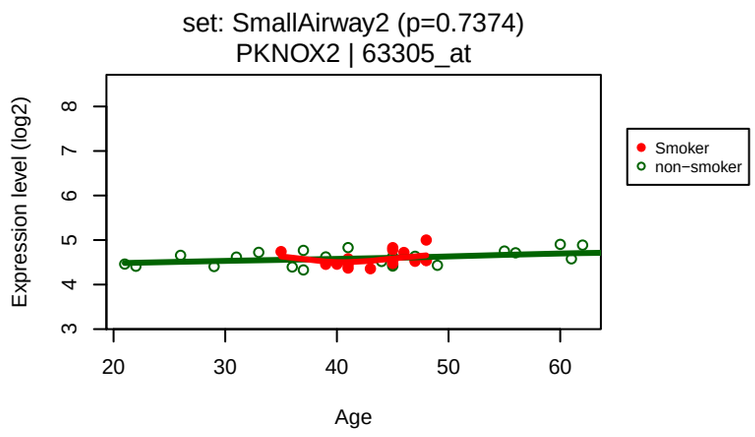
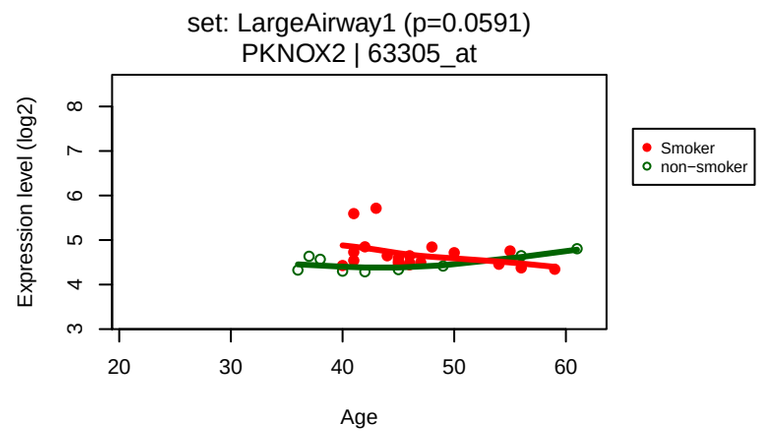

Gene expression of smokers and nonsmokers
gene: PKNOX2 | 63305_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2632
p=0.7374
p=0.2972
p=0.0591
p=0.5821
p=0.4257
p=0.6682
p=0.0659
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

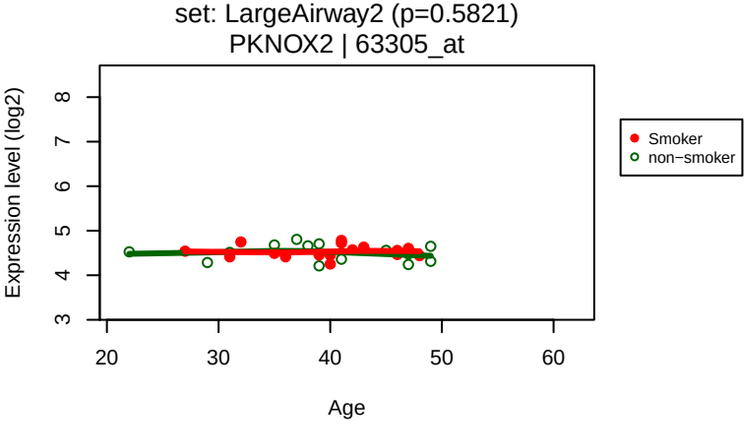

Gene expression ratio (smokers/nonsmokers)
gene: PKNOX2 | 63305_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2632
p=0.7374
p=0.2972
p=0.0591
p=0.5821
p=0.4257
p=0.6682
p=0.0659
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

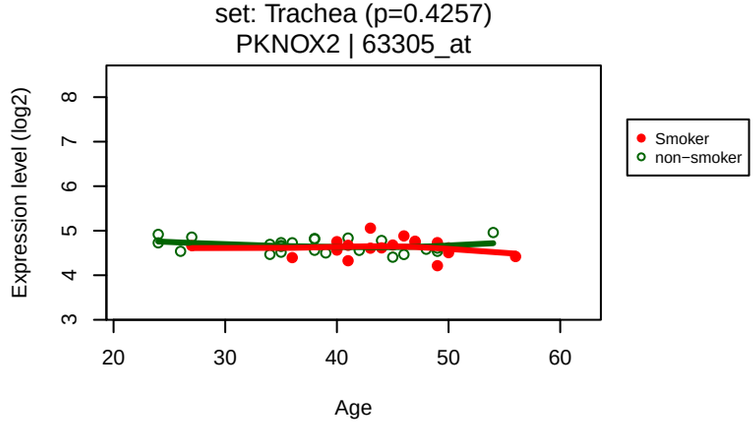

Gene expression ratio in Males
gene: PKNOX2 | 63305_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2473
p=0.2382
p=0.824
p=0.0313
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PKNOX2 | 63305_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5408
p=0.0237
p=0.0761
p=0.7029
SmallAirway2
LargeAirway2
Trachea
Macrophage2

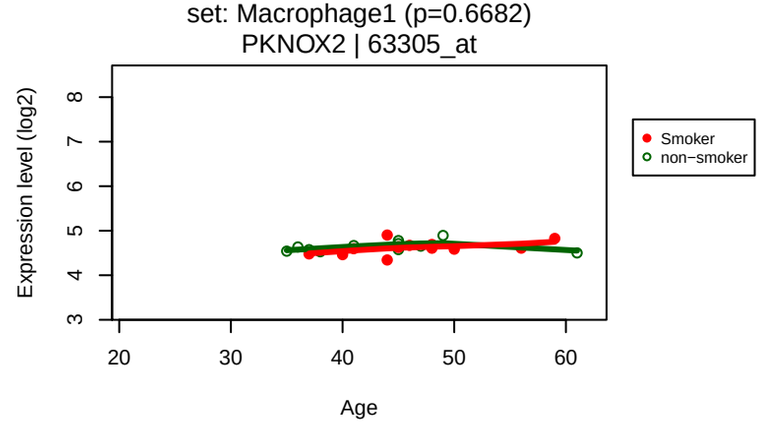

set: SmallAirway1 (p=0.2632)
PKNOX2 | 63305_at
Expression level (log2)
Age
Smoker
non−smoker

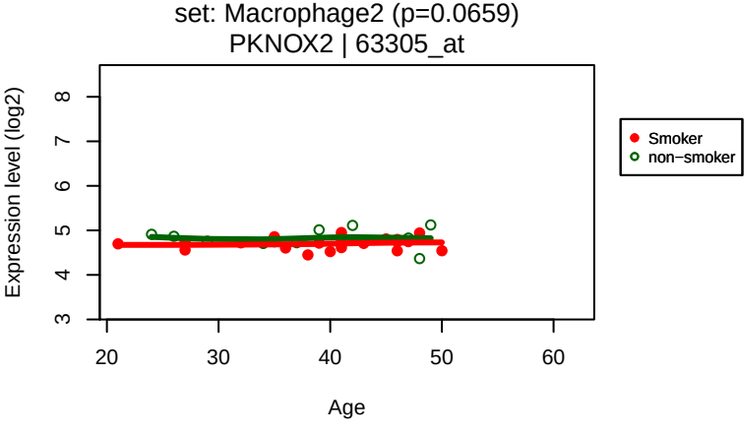

set: SmallAirway2 (p=0.7374)
PKNOX2 | 63305_at
Expression level (log2)
Age
Smoker
non−smoker

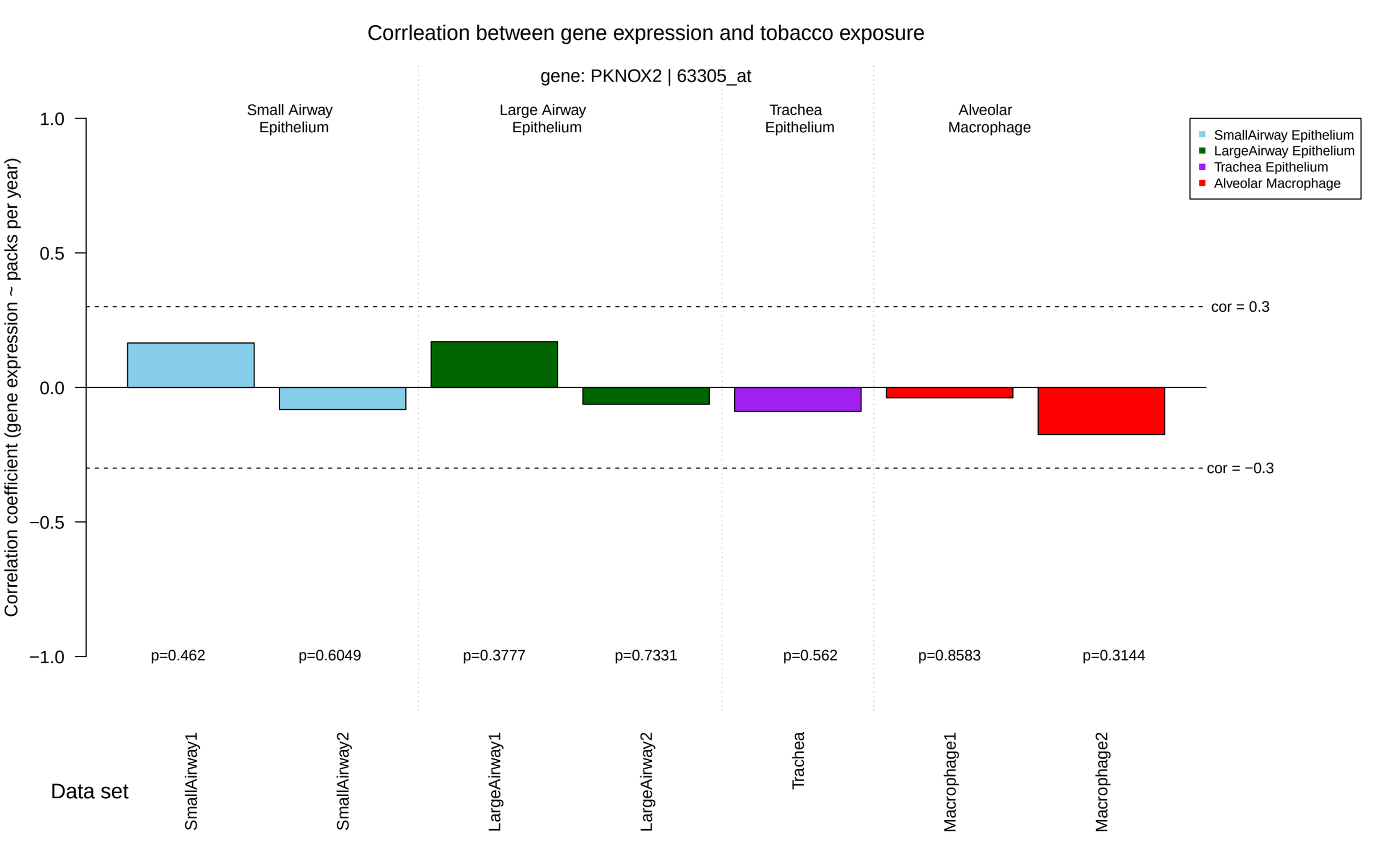

set: LargeAirway1 (p=0.0591)
PKNOX2 | 63305_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5821)
PKNOX2 | 63305_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4257)
PKNOX2 | 63305_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6682)
PKNOX2 | 63305_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0659)
PKNOX2 | 63305_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PKNOX2 | 63305_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.462
p=0.6049
p=0.3777
p=0.7331
p=0.562
p=0.8583
p=0.3144
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2